

Propositional eternalism, temporalism, necessitism and contingentism

Handout for *Time and Modality* seminar

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1 Recap

T1 Are there temporarily true propositions?

Propositional temporalism: yes. *Propositional eternalism*: no.

M1 Are there contingently true propositions?

Propositional contingentism: yes. *Propositional necessitism*: no.

2 Ground rules: truth and attitudes

Truth schema: The proposition that ϕ is such that: for it to be true is for it to be the case that ϕ

- What counts as an instance of the schema
- What status is claimed for instances of the schema: truth on all uniform interpretations

Belief Schema: The proposition that ϕ is such that to believe it is to believe that ϕ .

Assertion Schema: The proposition that ϕ is such that to assert it is to assert that ϕ .

Bad generalisation: The proposition that ϕ is such that to VERB it is to VERB that ϕ .

- Where VERB can be any verb capable of taking a clausal complement.

Better generalisation: The proposition that ϕ is such that to bear the “VERB-ing relation” to it is to VERB that ϕ .

- Works for stative VERBS. Need something more complicated for eventive VERBS like ‘claim’, verbs taking more arguments like ‘tell’, etc.

Can we do without the word ‘proposition’?

- A kind of quantification that isn’t obviously identical to quantification over propositions:
The pope told me that miracles sometimes occur.
Everything the pope told me is true.
Therefore it is true that miracles sometimes occur.
Therefore miracles sometimes occur.
We could use this to raise questions in the vicinity of T1 and M1, e.g. ‘Is everything that could be said and that is true always true?’ Or just ‘Is everything true always true?’
- Use quantification into sentence position?

$$\forall p(p \rightarrow \text{Always } p)$$

A variant view in the spirit of propositional eternalism/necessitism: temporal/modal qualifications cannot meaningfully be applied to ‘true’.

3 Some flat-footed arguments for contingentism, temporalism, “spatialism”...

- (1) Obama is president.
- (2) It is not necessary that Obama is president.
- (3) The proposition that Obama is president is such that for it to be true is for Obama to be

president

- (4) The proposition that Obama is president is such that necessarily: it is true iff Obama is president (from 3)
- (5) The proposition that Obama is president is true (from 1 and 4).
- (6) The proposition that Obama is president is not necessarily true (from 2 and 4).
- (7) Some proposition is true and not necessarily true (from 5 and 6).

Modal truth schema: The proposition that ϕ is such that necessarily, it is true iff ϕ .

Temporal truth schema: The proposition that ϕ is such that always, it is true iff ϕ

Spatial truth schema: The proposition that ϕ is such that everywhere, it is true iff ϕ

4 Bindability and the interpretation of schemas

A silly argument:

- (8) People admire him.
- (9) Some politician is sad because people do not admire him.
- (10) The proposition that people admire him is such that for it to be true is for people to admire him.
- (11) The proposition that people admire him is such that some politician is sad because it is not true.
- (12) The proposition that people admire him is true (from 1 and 4).
- (13) The proposition that people admire him is such that some politician is sad because it is not true. (from 2 and 3).
- (14) Some proposition is true and such that some politician is sad because it is not true (from 4 and 5).

Lesson: a propositional eternalist [necessitist] should think that occurrences of sentences under non-vacuous temporal [modal] operators cannot count as “uniformly interpreted” with respect to occurrences of the same sentence not under any temporal [modal] operator.

5 Is it just a verbal dispute?

A common thought: every true_1 proposition₁ is always true_1 , but some true_2 proposition₂ is not always true_2 .

If the truth schema holds on both interpretations, then we can put them together to get:

- (i) The proposition₁ that ϕ and the proposition₂ that ϕ are such that: for the former to be true_1 is for the latter to be true_2 .

Prima facie this entails:

- (ii) The proposition₁ that ϕ and the proposition₂ that ϕ are such that: always, the former is true_1 iff the latter is true_2 .

We could block this entailment by appealing to the view that material that is nonvacuously operated on by a tense operator can't be interpreted uniformly with respect to material that isn't: 'true₂' in (ii) just can't be read as uniform with 'true₂' in (i) unless it is read in such a way as to make the 'always' vacuous (if that's even linguistically possible).

But that view about tense operators doesn't seem to be neutral as between eternalists and temporalists: in fact it seems essentially to give the game away to the eternalist.

6 'Thank goodness that's over' and other psychological arguments

Some observations:

- (i) 'Annie thinks it is raining' seems false at noon in a circumstance where Annie has lost track of the time and thinks, at noon, that it is 3pm, and Annie thinks that it is sunny at 3pm (she is sitting in the sun) but that it was raining at noon.
 - Is 'Annie thinks it is raining now' different in this respect?
- (ii) 'Annie thought it was raining' is ambiguous. It has a "two-time" reading where it is true if Annie thought, in the past, that it was raining during some particular further-past interval. It also has a "simultaneous" reading where only one past time is relevant. On this reading, it's true if Annie was thinking to herself 'It is raining!' at that time, but it's not true if Annie had lost track of the time and thinking to herself 'It's not raining, but it was raining on such-and-such date at such-and-such time', referring to the time in question.
- (iii) 'Arthur is relieved that his dental surgery is over' seems consistent with 'There is no time t such that Arthur is relieved that his dental surgery is over by t '.

The "date theory":

When uttered at a certain instant i , a present-tense sentence ϕ expresses the same proposition that would be expressed by the sentence 'At t , ϕ ' relative to an assignment that maps the variable t to i .

How the foregoing facts pose a problem for the date theory.

Is there an analogous problem for propositional necessitists? (Tricky issues about what it takes to think at one possible world a thought about another one.)

7 Just another Frege puzzle?

Analogy: the theory ('Millianism'?) that 'Hesperus is bright' and 'Phosphorus is bright' express the same proposition faces the problem that 'Hammurabi believes that Hesperus is bright' seems to (potentially) come apart in truth value from 'Hammurabi believes that Phosphorus is bright'.

Responses:

- Pragmatic Millianism: 'Hammurabi doesn't believe that Phosphorus is bright' is literally false but acceptable for some other reason.
- Error-theoretic Millianism: 'Hammurabi doesn't believe that Phosphorus is bright' seems acceptable just because we wrongly believe that false proposition it literally expresses.
- Contextualism: 'believes' introduces context-sensitivity; the choice of names affects how this context-sensitivity is most naturally resolved.

How do the analogues of these responses look for the date theorist?

Propositional eternalism without the date theory? a Fregean approach

- For each time t , there is a proposition that is (i) necessarily, true iff it is raining at t , and (ii) necessarily, believed at t only by those who believe that it is then raining under a "present-tensed mode of presentation" (and similarly for other attitudes).
 - What, if anything, does it take to believe this proposition at a different time?

- When we utter a present tense sentence, the proposition expressed is of this sort. Similarly these propositions are designated by present-tensed 'that' clauses.
- In a past-tensed attitude report on the "simultaneous" reading, the 'that' clause designates a proposition of this sort.

8 'Centred propositions'

Lewis's concept of *self-ascription*.

- For Lewis: a relation between person-stages and properties that could be had by person-stages.
- A more neutral version: a relation between people, times, and functions from people and times to eternal propositions.
- Note of caution: 'John believes that his pants are on fire' certainly has *a* reading on which it can be true even if John doesn't self-ascribe *having pants that are on fire*.
 - It's controversial whether it has any reading that *does* require this.
 - If it does have such a reading, it's controversial whether it involves a different resolution of structural syntactic ambiguity.
- Lewis's account of "*de re* belief" in terms of self-ascription.

Lewis's analysis of in terms of self-ascription